CMAI 480
Computer Vision

Syllabus and Course Admin



Dr. Khalil Yousef

Department of Computer Science & Engineering

Qatar University

About the Instructor

- **Dr. Khalil Yousef**
  - **Office: H07 - C254**
  - **Office hours:**
    - **Female Sections: Sunday, 12 pm – 12:50 pm, H07 – A399**
    - **Male Sections: Tuesday, 1 pm – 1:50 pm, H07 – B114**
    - **Additionally: You can talk to me after the class if you have quick issues/questions**
  - **Best way to contact me is by Email**

kyousef@qu.edu.qa

Course Learning Outcomes (CLO's)

1. Understand and master basic knowledge, terminology, mathematics, models and methods in the field of computer vision.
2. Identify, formulate and solve problems in computer vision through appropriate assessment and selection from the existing methods.
3. Design and develop practical and innovative image processing and computer vision applications or systems.

Course Prerequisite

- CMPS 251: OOP

- Data structures, C++/Python programming, Basics of mathematics

Your Grade is Based on:

Programming Assignments	15%	[4 Assignments]*
Project	35%	
Midterm Exam	20%	**Week 7**
Final exam	30%	

***No assignment make-up will be given**

Course Project

- A list of suggested topics will be shared with you soon on Blackboard.

- Teams of 5 students

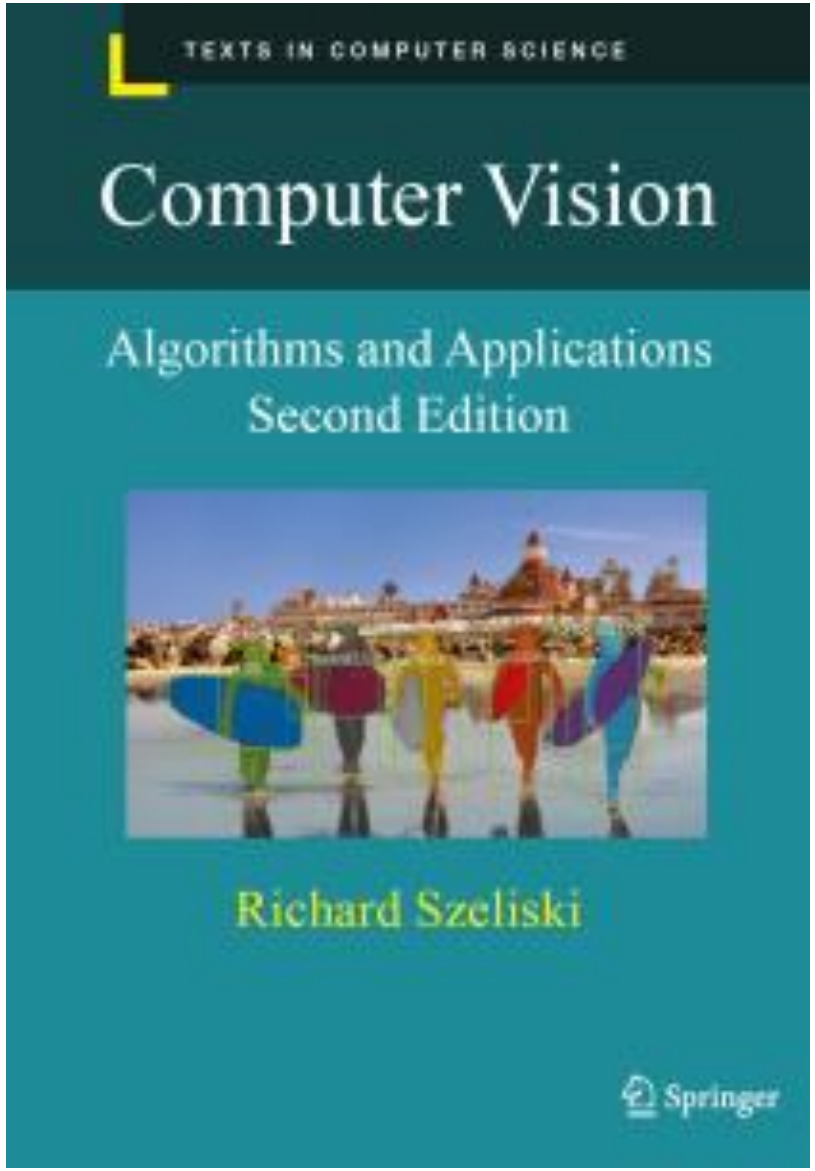
Course Topics and Tentative Schedule

Tentative Course Schedule

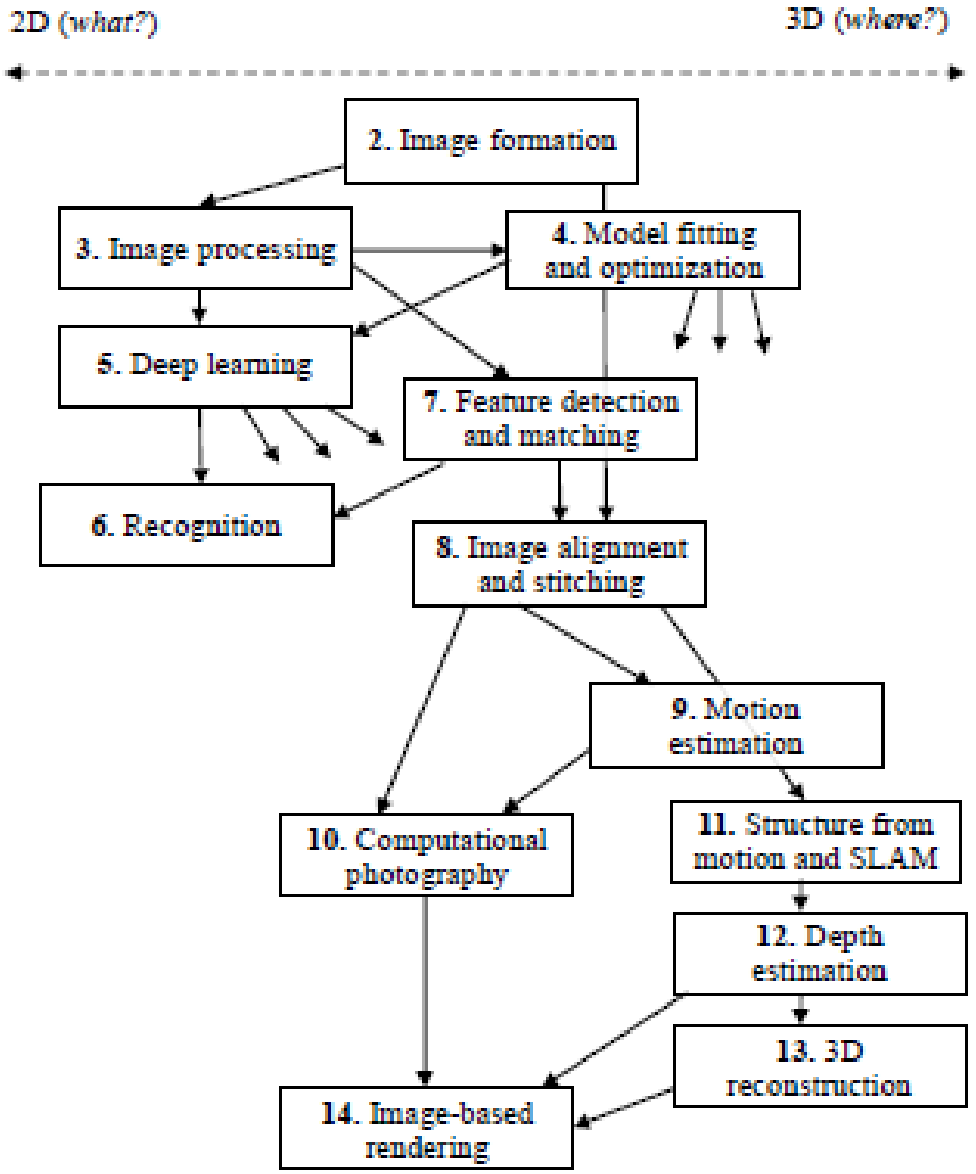
	Topics	Chapter/Section(s)	Weeks
23/08/26 W 1	Introduction to Computer Vision (CV)	1.1	1
30/08/26 W 2&3&4	Image Formation + Assignment (1) – Due End of W3	2.1,2.2,2.3	3
20/09/26 W5	Image Processing + Assignment (2) – Due End of W5	3	1
27/09/26 W 6 &7	Feature Extraction and Matching + Assignment (3) – Due End of W7 Midterm Exam – W 7 (includes all course material covered up to (**but not including**) Feature Extraction and Matching)	7	2
11/10/26 W8&9	Image Alignment and Stitching + Assignment (4) – Due End of W9	8	2
25/10/26 W 10	Mid-Fall Break	-	1
01/11/26 W 11	Introduction about camera calibration	11	1
8/11/26 W 12 & 13	Fundamentals of machine-learning and Deep Learning on Course Project: Classification and Semantic Segmentation	4,5	2
22/11/26 W 14	Projects Presentations		1
29/11/226 W 15	Dead Week - Review (Last day of classes - 3 December 2026)		1
Total Weeks			15

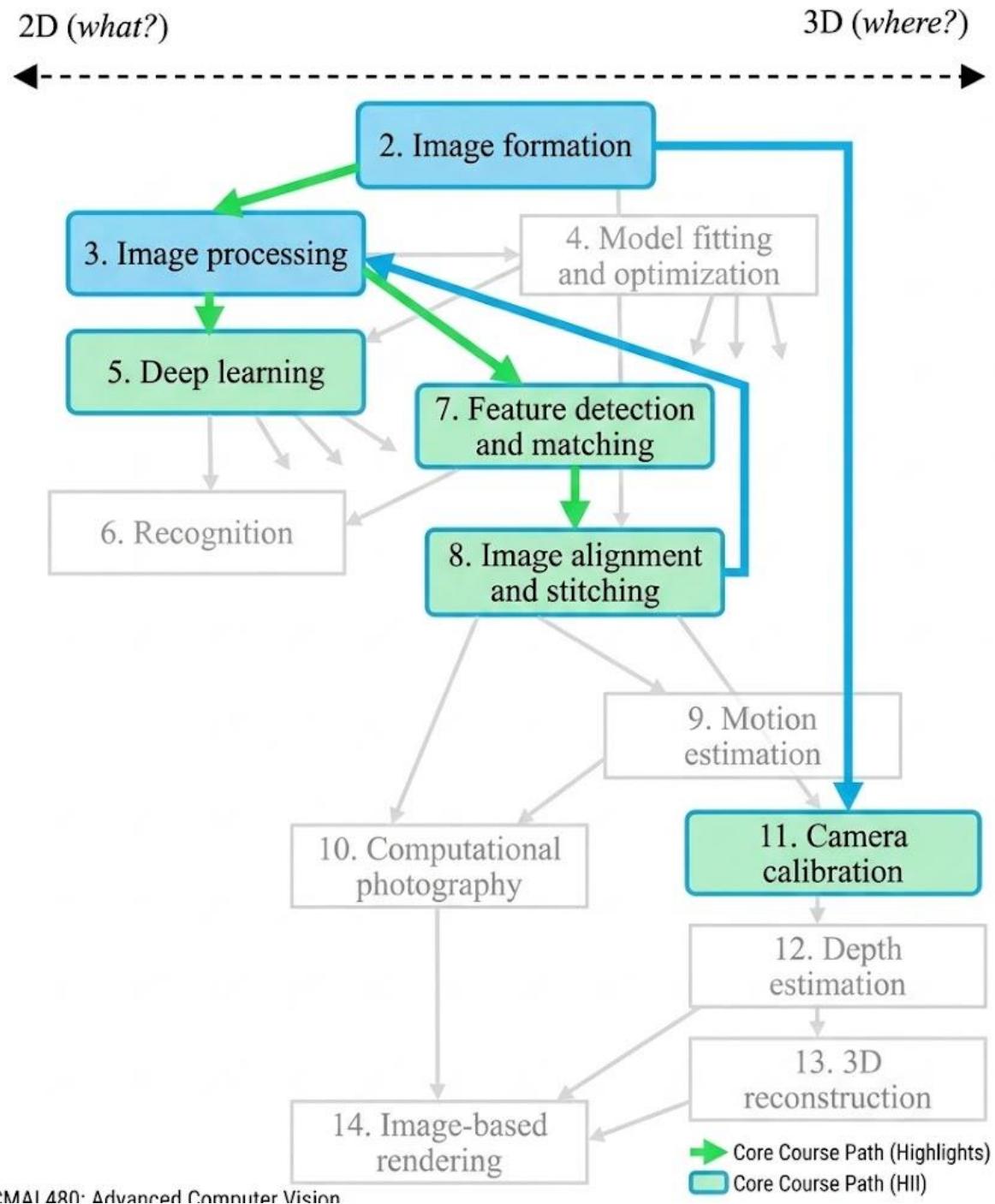
The Textbook

- Computer Vision: Algorithms and Applications" by Richard Szeliski which can also be found online: http://szeliski.org/Book



The Textbook: Taxonomy of Topics





Note: HII / High-Impact Interconnections (Blue Arrows): Highlights **cross-topic dependencies or direct applications**

Learning Progression

1. Fundamentals

Image formation, digital sensors, and 2D spatial filtering techniques.

2. Feature Extraction

Corner detection, SIFT descriptors, and pairwise point matching.

3. Geometric Alignment

Homographies, camera calibration matrices, and image panoramas.

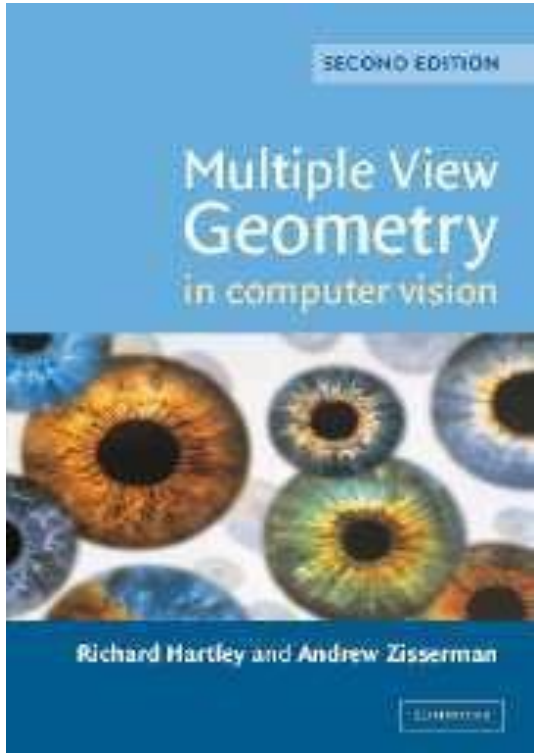
4. Deep Learning

End-to-end neural networks for image recognition and vision tasks.

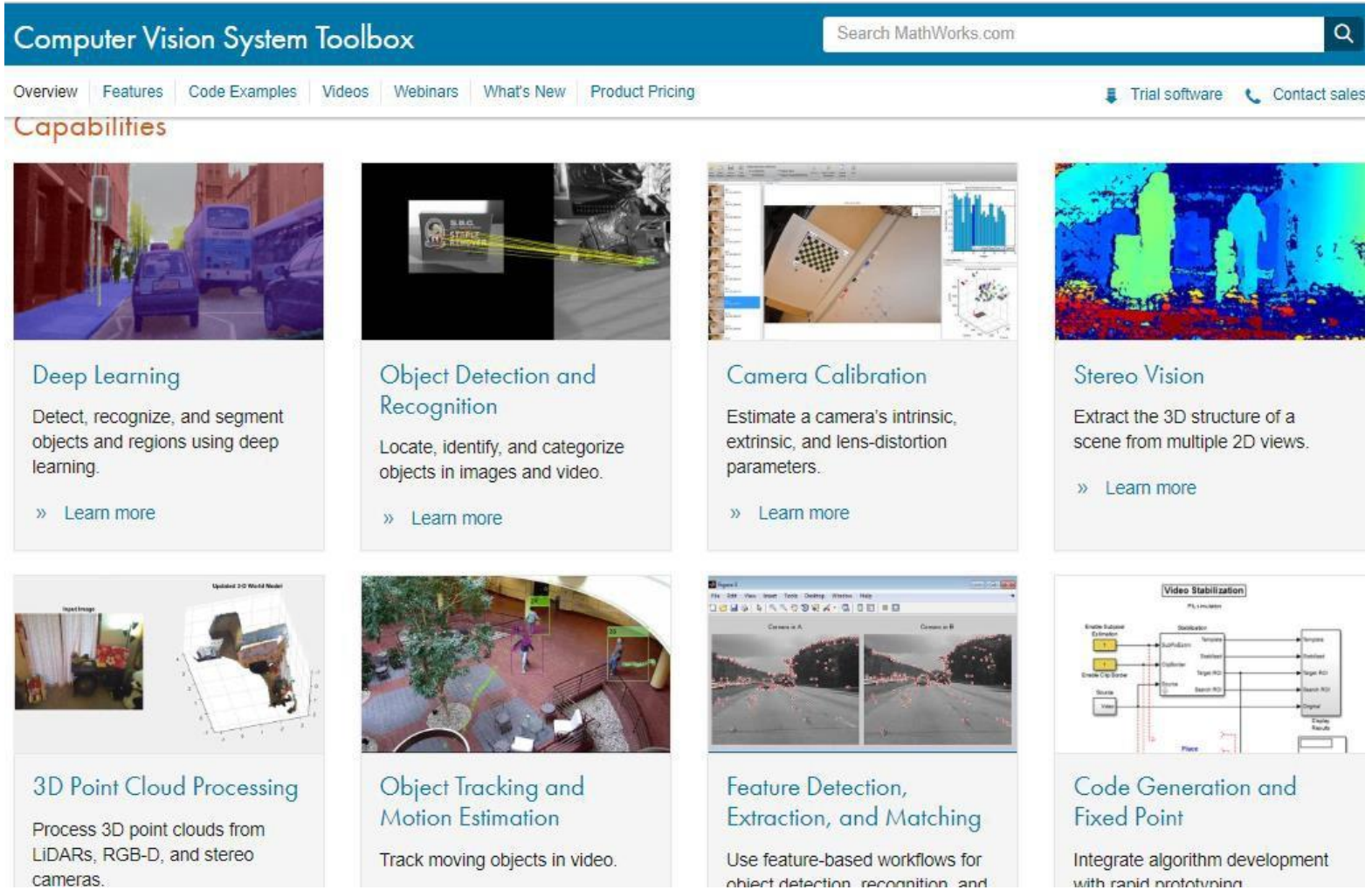
Reference

- Multiple View Geometry in Computer Vision, 2nd Edition, Richard Hartley, Andrew Zisserman, published by Cambridge University Press, 2003.

- An excellent reference you should own
  - http://www.robots.ox.ac.uk/~vgg/hzbook/
  - Matlab Code
    - http://www.robots.ox.ac.uk/~vgg/hzbook/code

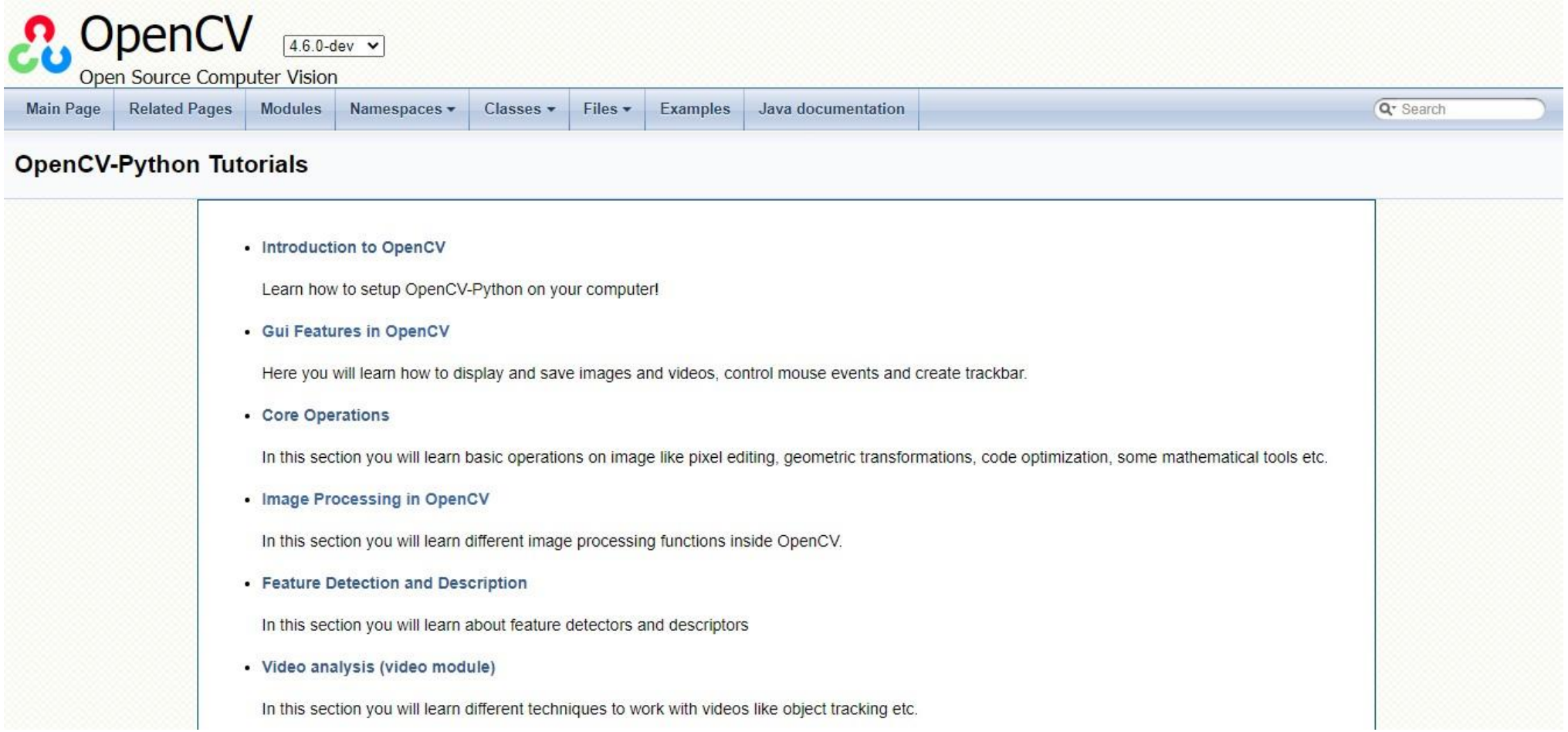


Other References: Matlab Computer Vision Toolbox



https://www.mathworks.com/products/computer-vision.html

OpenCV-Python Tutorials



https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html

Software Usage

- Matlab
  - https://www.mathworks.com/academia/tah-portal/qatar-university-31688665.html

Attendance Policy

- Attendance is mandatory according to university policies and more than 25% absence will not qualify you for the course credit (11 absences).

Plagiarism / Cheating

- “Getting an unfair academic advantage"
  - Using other people's work as your own
  - Not doing your assignments yourself
- All the code you submit has to be your own
  - **NO** code you have found on the web or that being generated by an AI. **NO** sharing with others.
- **Do your assignment and project yourself**
  - Do NOT copy from each other or from the Internet
  - You can be picked-up randomly to explain your implementation
  - Cheating will be treated very seriously
- Penalties START with a zero on the assignment, failing the course! and other disciplinary actions as per QU policy

Plagiarism / Cheating

- In accordance with Article 6 of the Student Code of Conduct at Qatar University, academic violations include a range of actions, one of which pertains to submitting work that is not the individual's own production. This includes using creative artificial intelligence tools such as **ChatGPT** to produce content, images, videos, or programming code and presenting it as original work. Therefore, students are cautioned that using artificial intelligence tools such as **ChatGPT** or any similar tools to produce academic content and present it as their own work is considered plagiarism, exposing the student to disciplinary penalties as stipulated in Qatar University's Student Code of Conduct.
- In light of this, we urge all students to adhere to ethical standards in all assignments and academic work, and to seek guidance from the course instructor when unsure about the proper and ethical use of artificial intelligence sources in completing assignments, duties, and academic tasks.

No ‘Free Riding’ allowed

- ‘free riders’ (who do not contribute much) => not acceptable and not fair for hardworking students
  - You must actively contribute to your project and do your ultimate best to deliver the best possible results
  - Otherwise you will be asked to do the project alone



Email Rules

- When emailing me you must add – **CMAI480** to the beginning of the email title and your section number
  - e.g., CMAI480 – L01 – Request for a meeting
  - e.g., CMAI480 – L51 – Regrade Request
  - e.g., CMAI480 – L52 – Question Regarding … Topic

Learning Support Outside of Class

- Myself
- Your Colleagues
- Online Resources

Questions

Thank You!